

An illustration in the top left corner shows a person in a blue silhouette sitting at a desk, talking on a phone. Above them are icons of a gear and a lightbulb, symbolizing thought and technology.

# Simple Linear Regression

# Course Objectives

What is Simple Linear Regression?

The activities involved in building

How to validate a Linear Regression Model?

# Learning Outcomes

At the end of the course, you will understand the concept of

- building a simple linear regression model using past data
- making predictions using the model
- determining the accuracy of your predictions

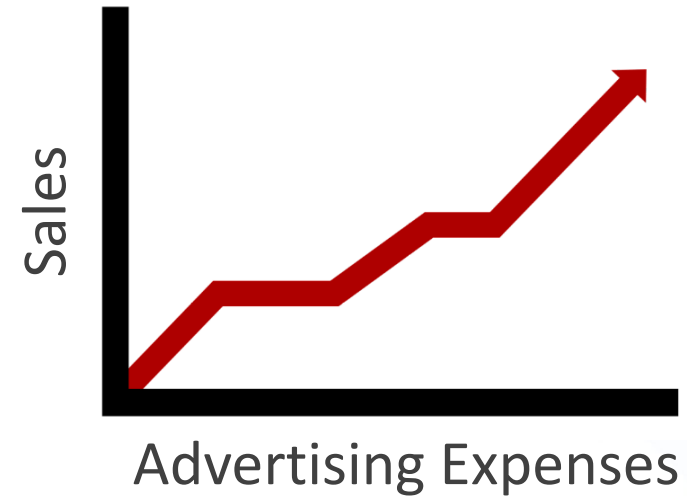
# Linear Regression - Example



# Linear Regression - Example

Problem  
Statement

How much sales can be  
generated from a given  
advertising budget?



# Machine Learning

## Task – Sales Forecast

|   | A      | B     |
|---|--------|-------|
| 1 | Advert | Sales |
| 2 | 1000   | 10000 |
| 3 | 2000   | 19000 |
| 4 | 3000   | 29000 |
| 5 | 4000   | 46000 |
| 6 | 5000   | 53000 |
| 7 | 6000   | 67000 |
| 8 | 7000   | 71110 |

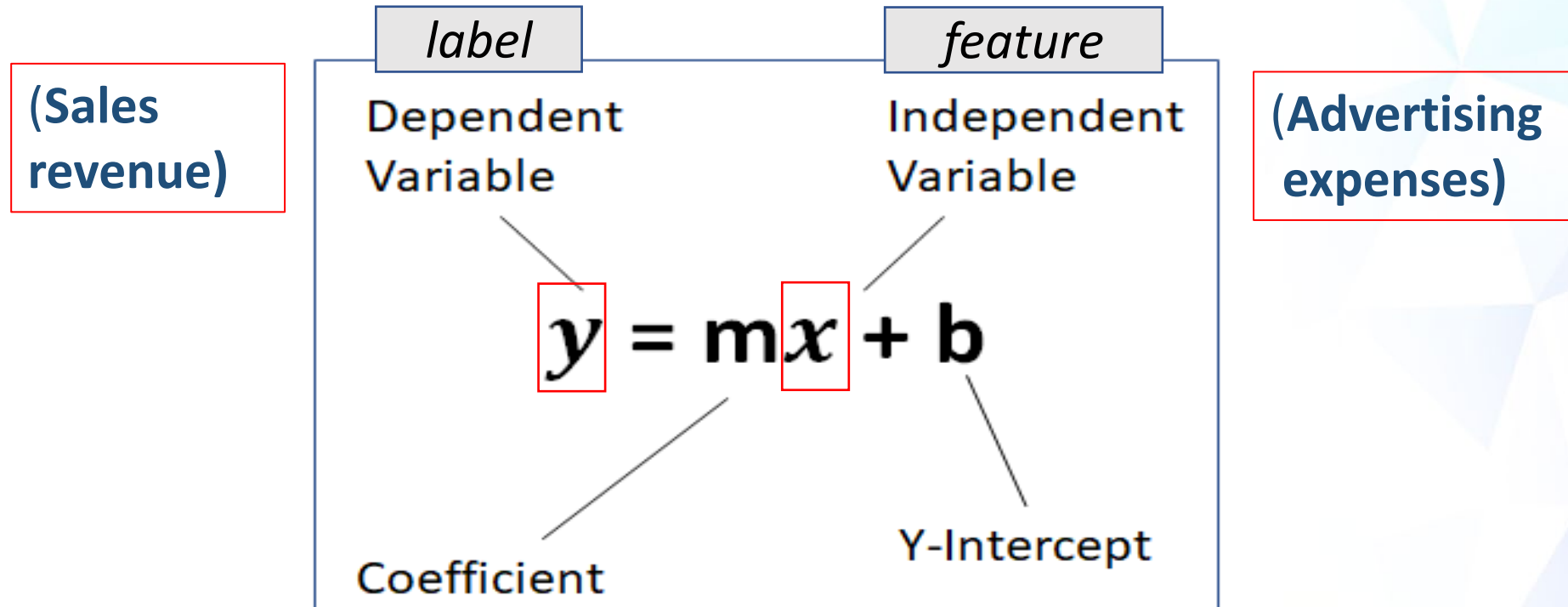
X  
(feature)

Y  
(label)

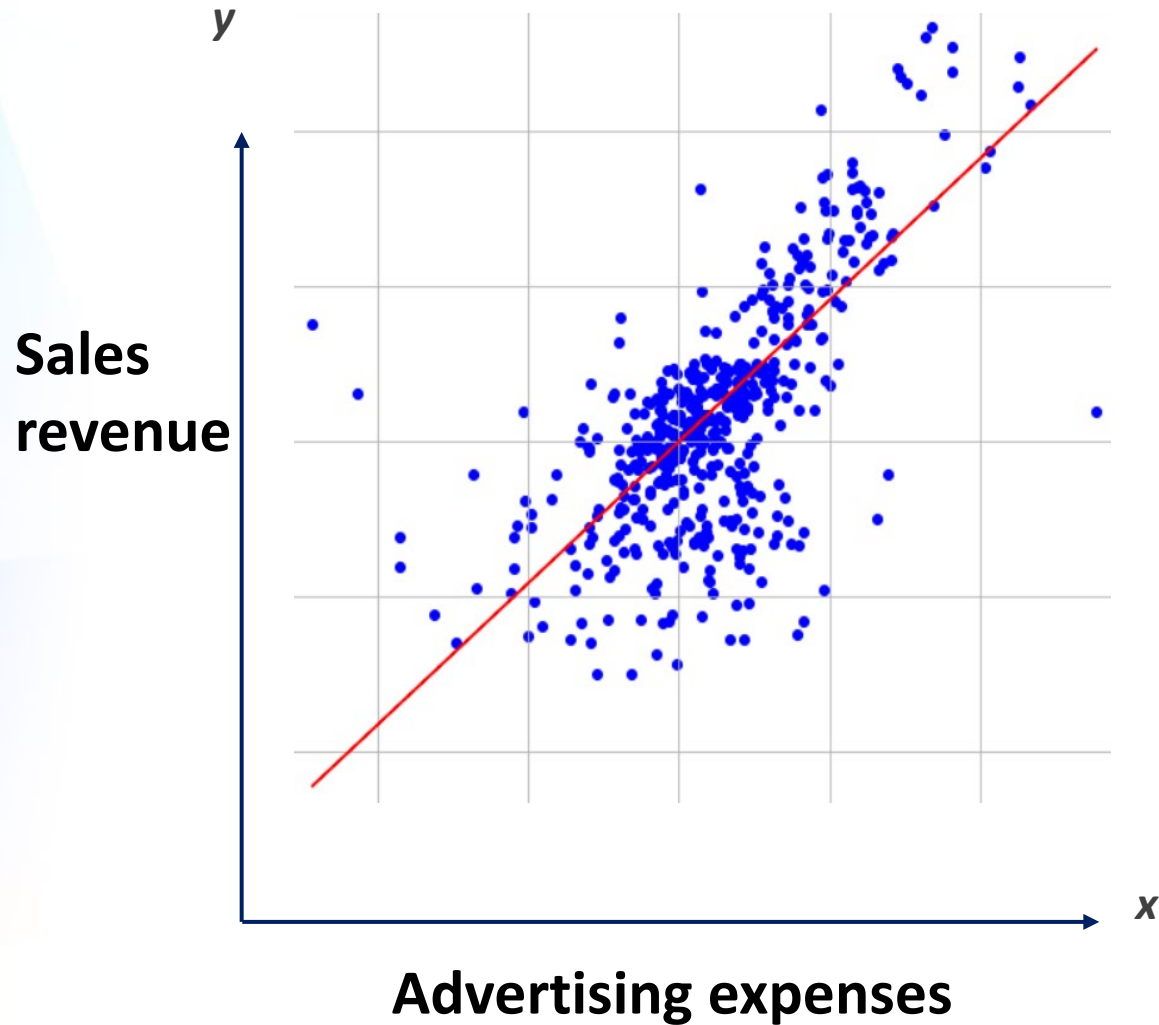
  
 Send the data
   
*Linear  
Regression  
Model*

# Machine Learning – Linear Regression

## Mathematical Formula

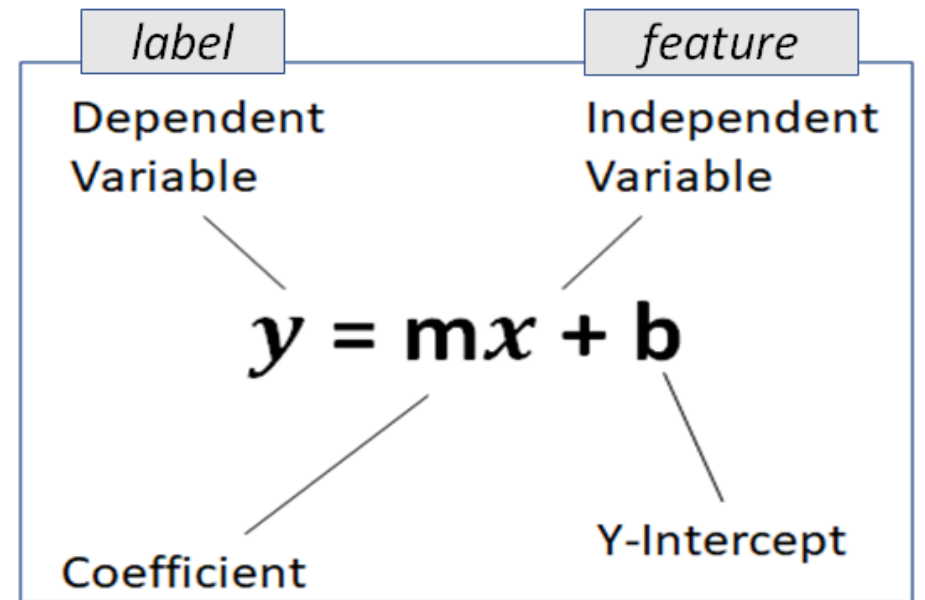


# Machine Learning – Linear Regression



Linear  
Regression  
Line

Supporting Mathematical Formula

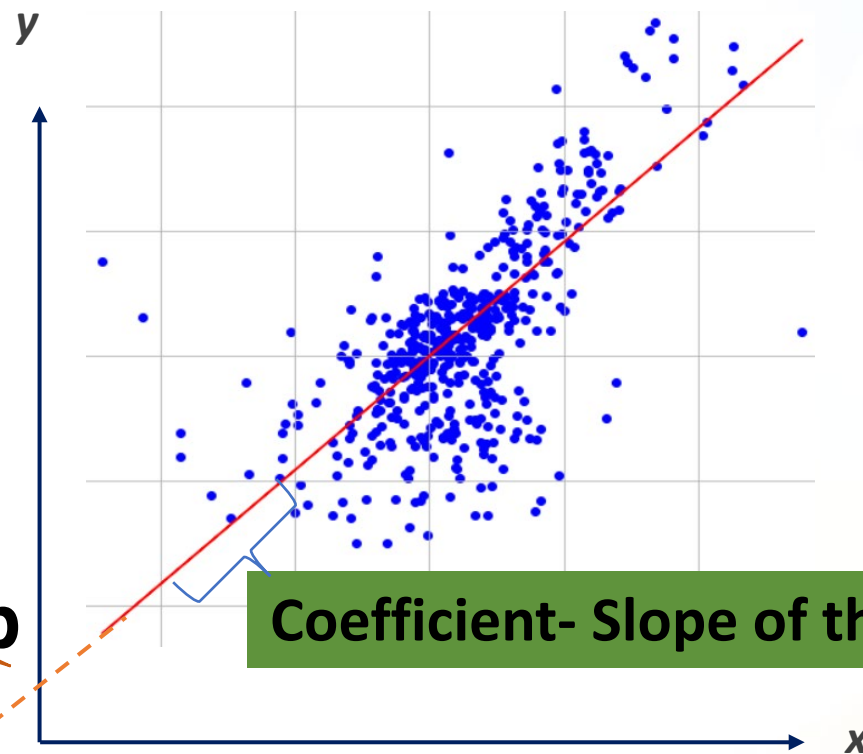
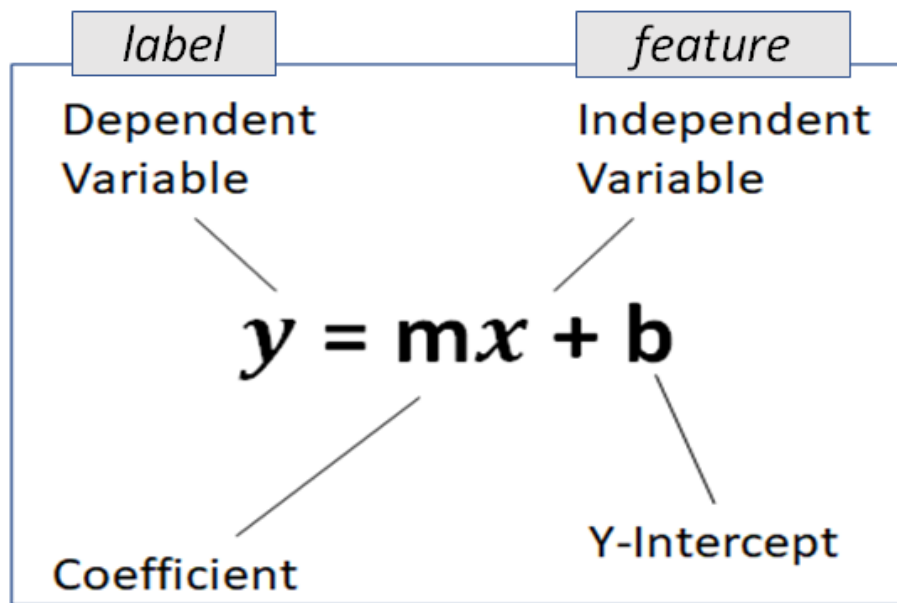




# Machine Learning – Linear Regression

## Supporting Mathematical Formula

Linear  
Regression  
Line



Y-intercept

$b$

Coefficient- Slope of the line

$m$

# Linear Regression Model

|   | A      | B     |
|---|--------|-------|
| 1 | Advert | Sales |
| 2 | 1000   | 10000 |
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| 8 | 7000   | 71110 |

*Problem Statement*

*Prepared Data*

How much sales can be generated from a given advertising budget?

*train*

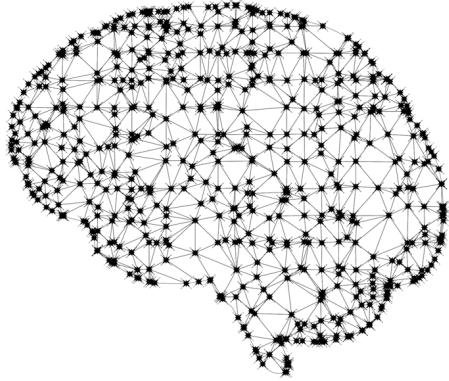
**Model**

Advertising budget =  $x$

Sale forecast,  $y$ ?

*Start Prediction!*

# Validating the Model



Trained ML model

The  $R^2$  of the model is 0.8! There is a strong correlation between advertisement expenses and sales.

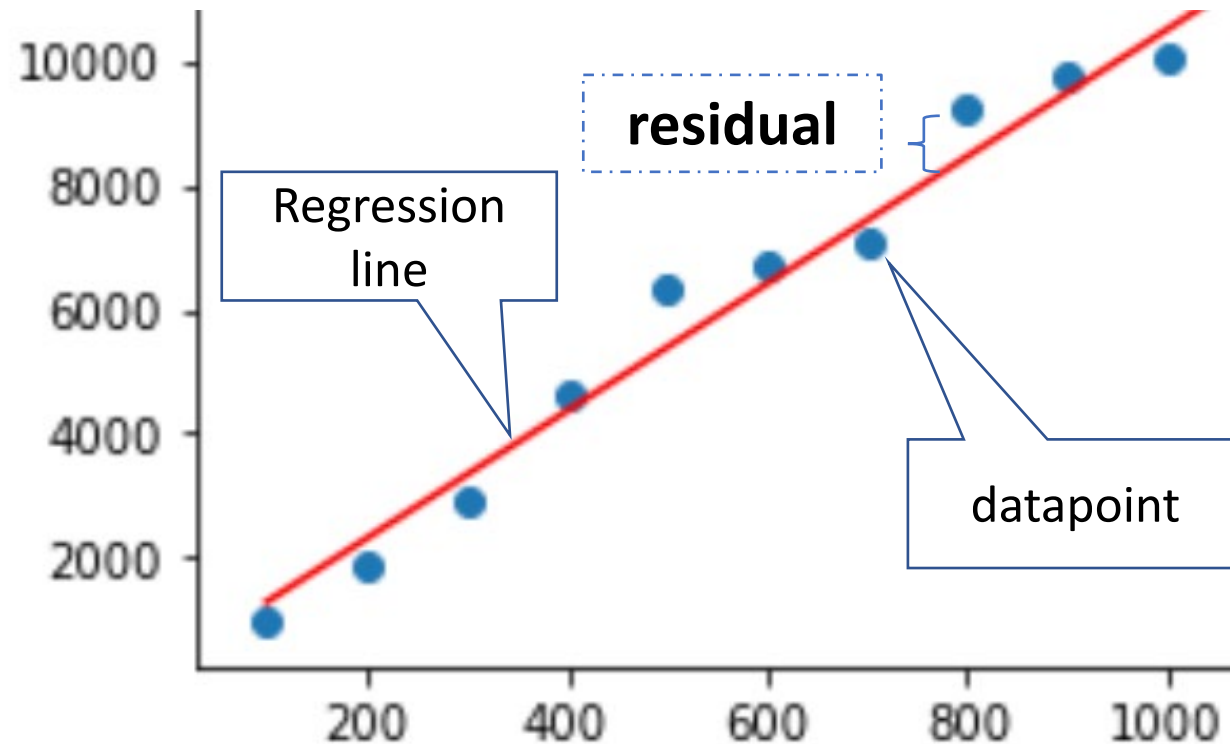


# How to validate the model

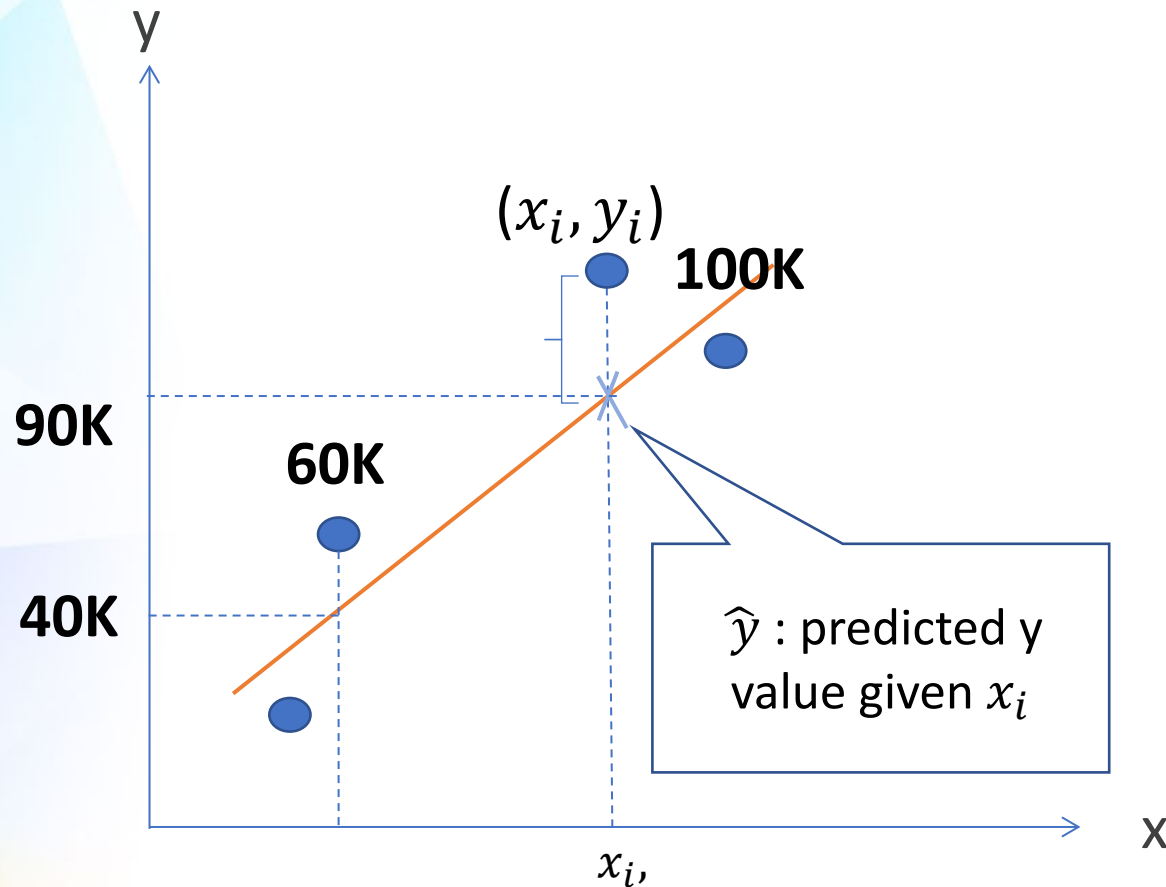
- 2 common types of validation
  - Mean Square Error
    - MSE is the average of the squares of the differences between the actual values and the predicted values.
  - R-Squared Score
    - measures the proportion of the variance in the dependent variable ( $Y$ ) that is predictable from the independent variable(s) ( $X_1..X_n$ ).
    - It indicates how well the data points fit the regression line.

# What are Residuals

- Residual is the ***difference between the actual and the predicted*** value of a regression model.
- We will use this concept to derive ***Mean Square Error*** and ***R-Squared***.



# Mean Square Error



Sum of Squared Errors (SSError): This is the sum of the squares of the differences between the actual values and the predicted values. It is also known as *Sum of square of residuals*

$$\begin{aligned} \text{SSError} &= \sum (y_i - \hat{y})^2 \\ &= (100\text{k} - 90\text{k})^2 + (60\text{k} - 40\text{k})^2 \\ &\quad + \dots \end{aligned}$$

$$\begin{aligned} \text{Mean Square Error} &= 1/n \sum (y_i - \hat{y})^2 \\ &= \text{SSError}/n \end{aligned}$$

(Sum of Square Error (SSE) divided by number of observations  $n$ )

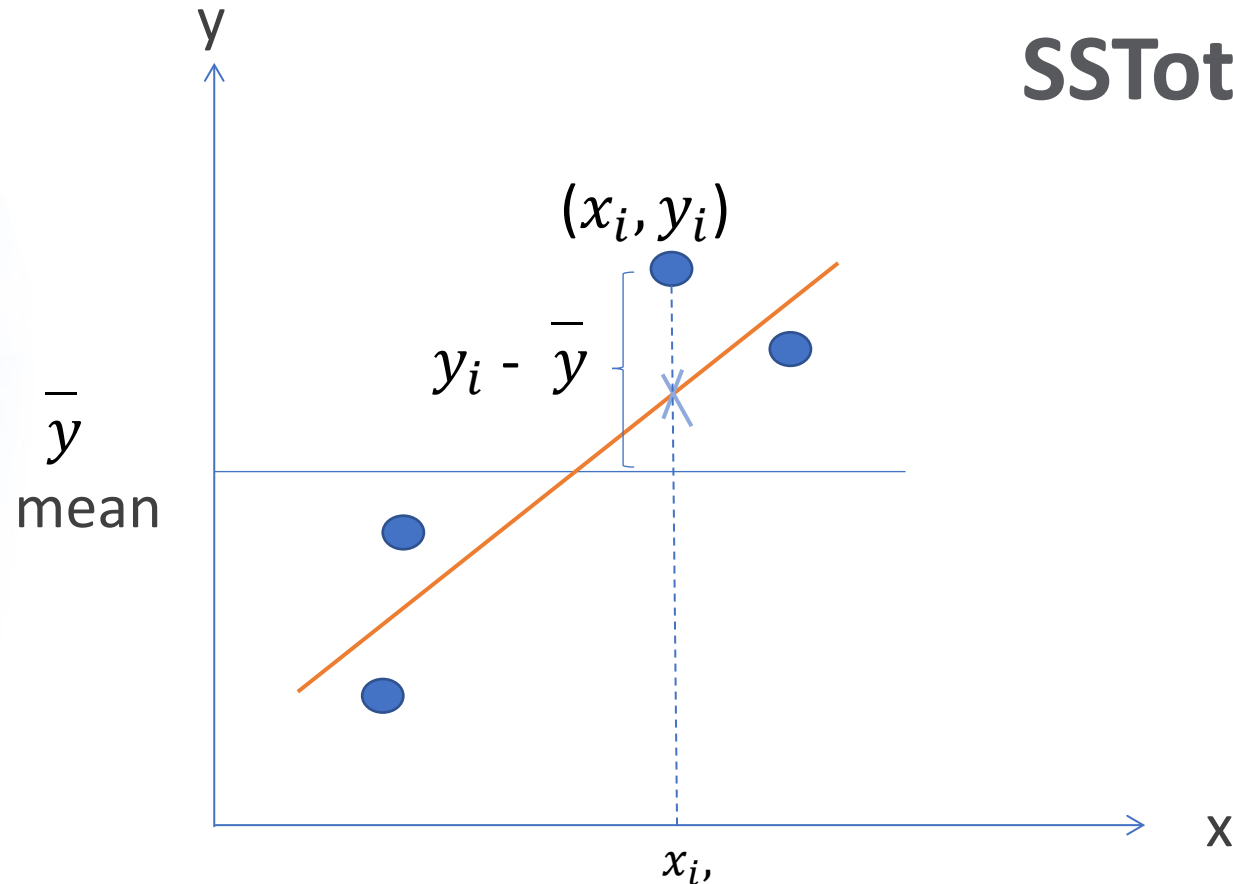
# Mean Square Error - Summary

- Indicates how close the predicted values are to the actual values.
- Lower MSE values indicate a better fit.
- Sensitive to outliers and can lead to a misleading conclusion.

# Total Sum of Squares

- Sum of the squares of the differences between the actual values and the mean of the actual values.

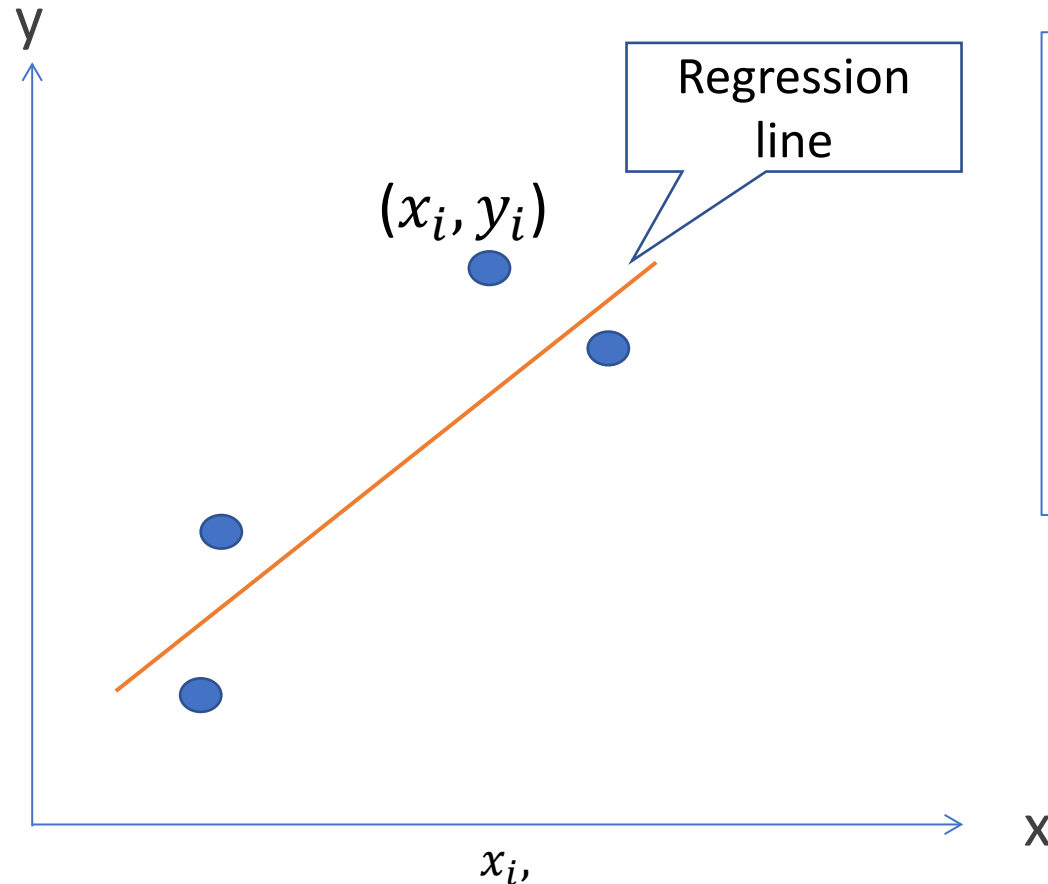
$$SSTotal = \sum (y_i - \bar{y})^2$$





# R-Squared ( $R^2$ )

- **R-Squared\*** - proportion of the variance in the dependent variable (y) that is predictable from the independent variables(x)



$$SSTotal = \sum (y_i - \bar{y})^2$$

$$SSError = \sum (y_i - \hat{y})^2$$

$$R^2 = 1 - \frac{SSError}{SSTotal}$$

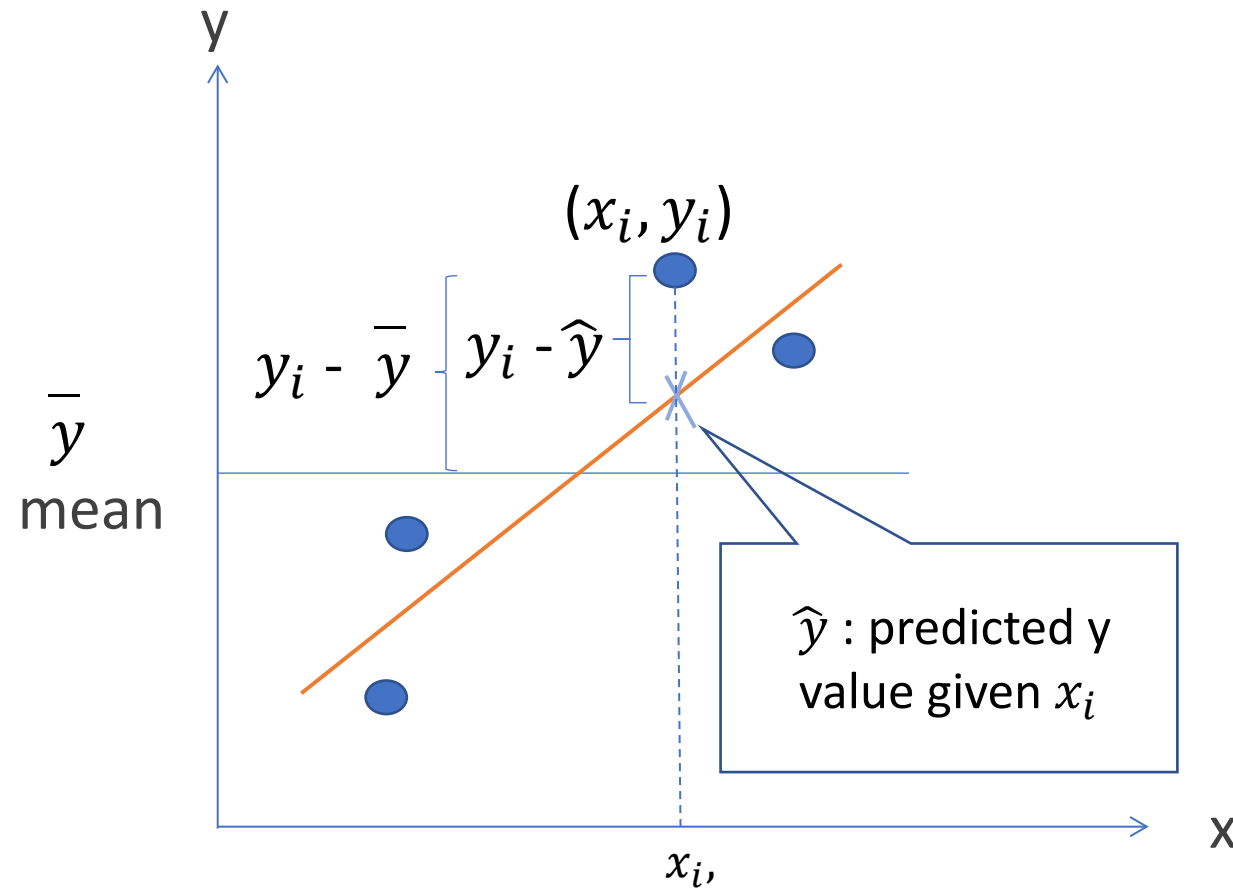
# $R^2$ Formula

When residuals are small,  $R^2$  will be close to 1.

$$\begin{aligned} SSTotal &= \sum (y_i - \bar{y})^2 \\ SSEerror &= \sum (y_i - \hat{y})^2 \end{aligned}$$

$$\begin{aligned} R^2 &= \frac{SSTotal - SSEerror}{SSTotal} \\ &= 1 - \frac{SSEerror}{SSTotal} \end{aligned}$$

# Value of $R^2$



$$SSTotal = \sum (y_i - \bar{y})^2$$

$$SSError = \sum (y_i - \hat{y})^2$$

$$R^2 = \frac{SSTotal - SSError}{SSTotal}$$

$$= 1 - \frac{SSError}{SSTotal}$$

$$= 1 - 0.04$$

$$= \mathbf{0.96}$$

# R-Squared - Summary

- Indicates the strength and direction of a linear relationship.
- Values range from 0 to 1, where 1 means perfect prediction.
- $R^2$  can increase simply by adding more independent variables to the model including those not having meaningful relationship with dependent variable.
- This can lead to overfitting, where the model fits the training data well but performs poorly on new data.

# Building a Linear Regression Model

*Problem Statement*

How much sales can be generated from a given advertising budget?

|   | A      | B     |
|---|--------|-------|
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*Prepared Data*

*train*

*Validate*

**Model**

$$R^2 = 1 - \frac{SSE_{\text{Error}}}{SS_{\text{Total}}}$$

*Start Prediction!*

# What Have We Learnt?

- What is a simple linear regression model
- The activities involved
- Validation of the model

$$y = mx + b$$

*Problem Statement*

*Prepared Data*

↓ *train*

*Validate*

**Model**

$$R^2 = 1 - \frac{SSE_{\text{Error}}}{SST_{\text{Total}}}$$

*Start Prediction!*